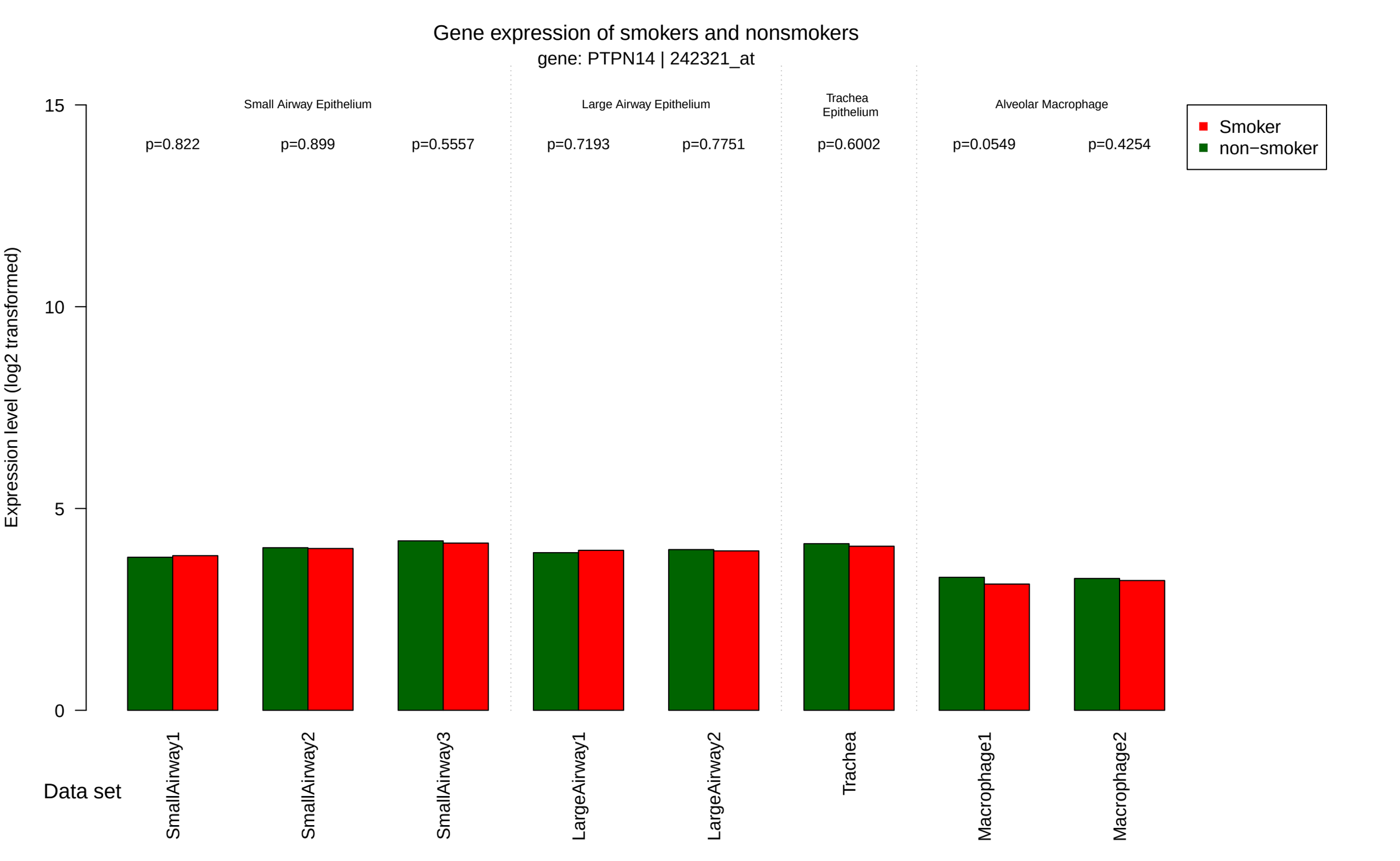
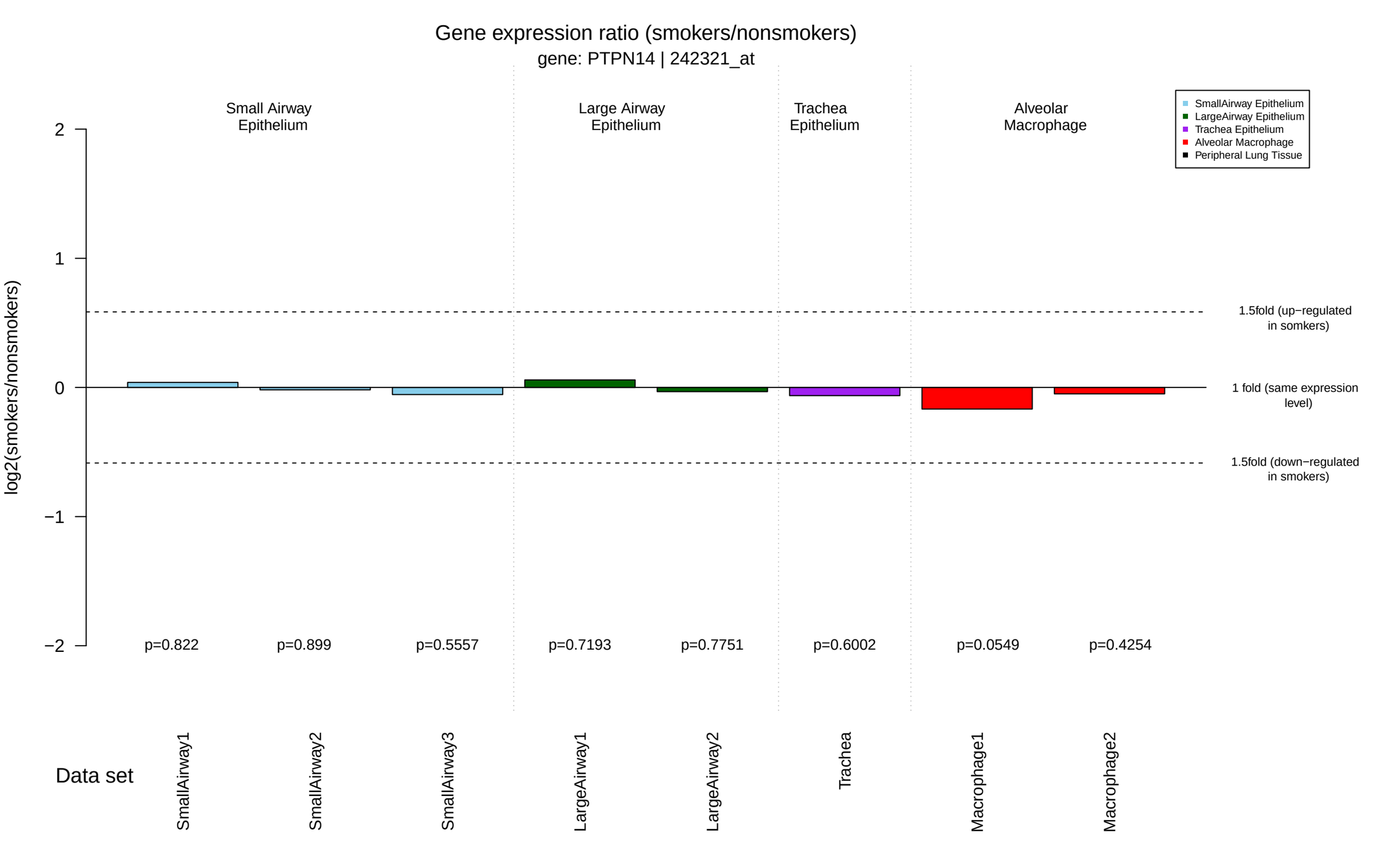
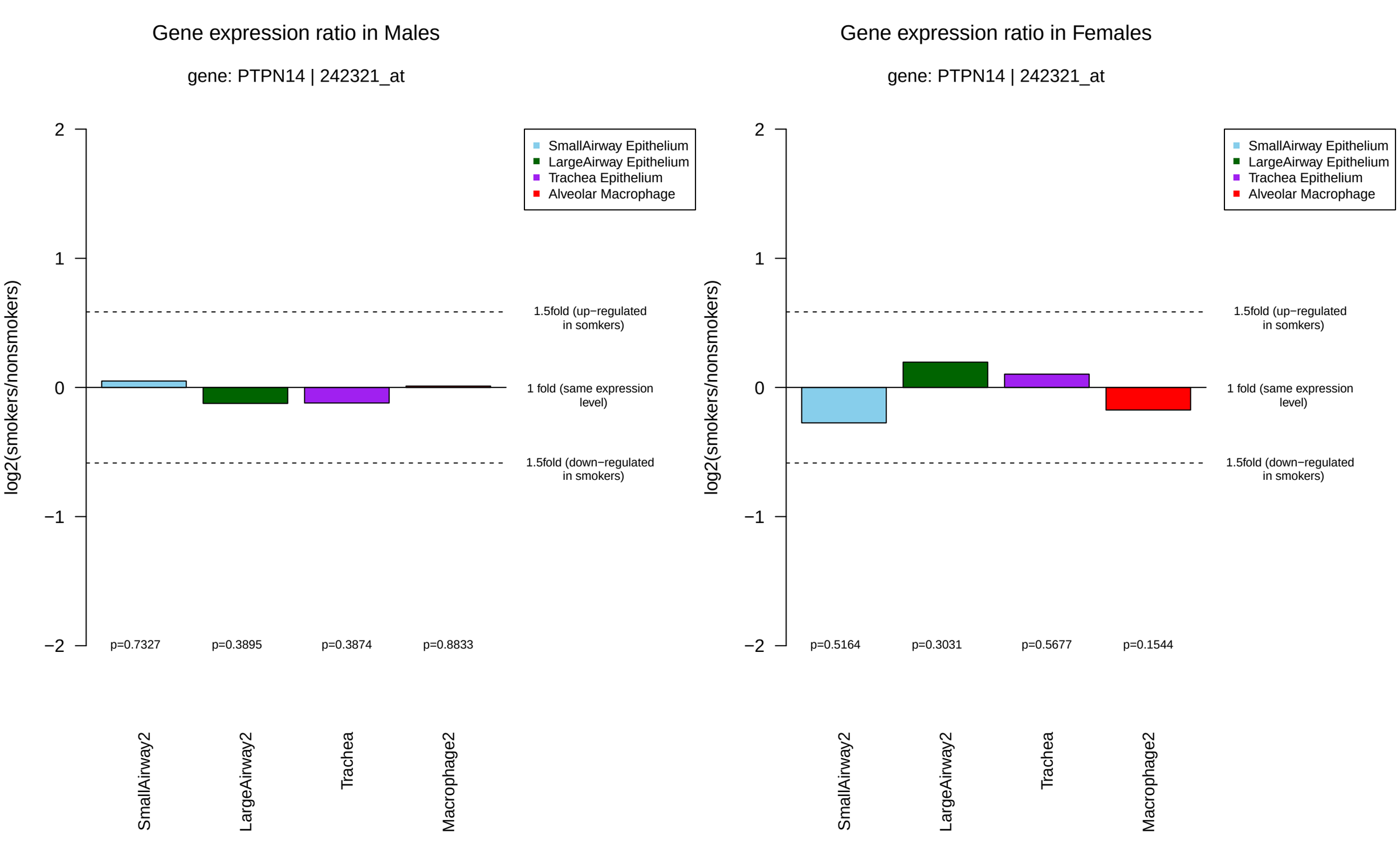
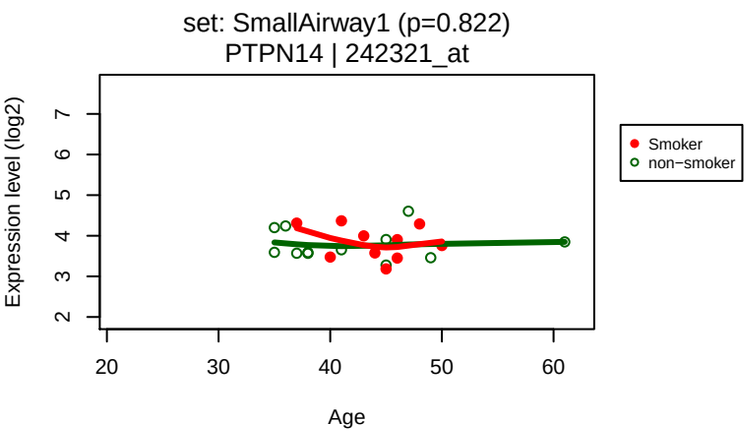
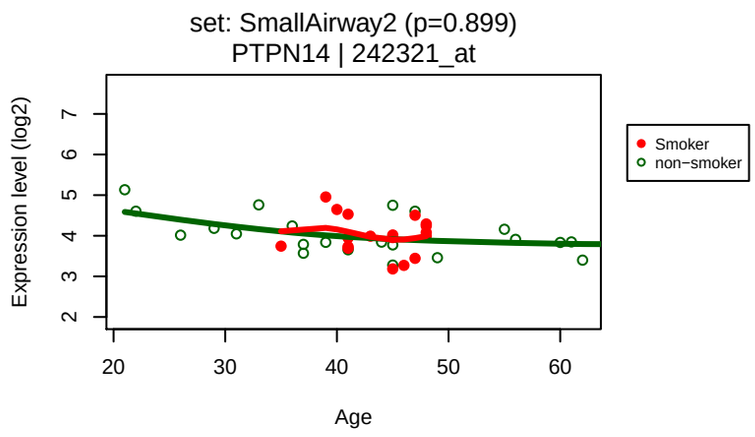
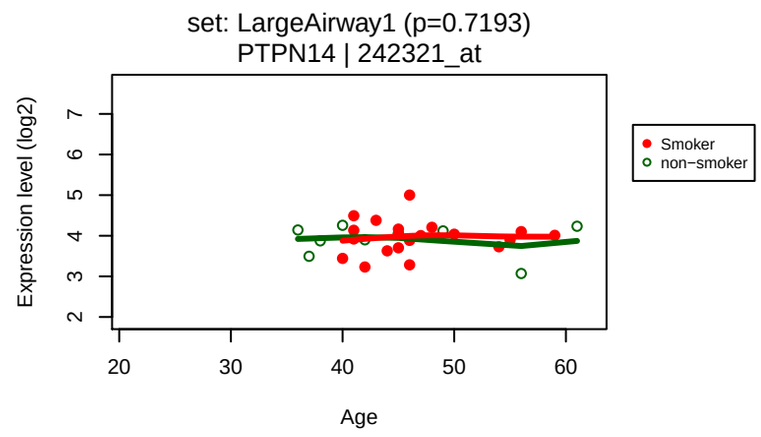

Gene expression of smokers and nonsmokers
gene: PTPN14 | 242321_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.822
p=0.899
p=0.5557
p=0.7193
p=0.7751
p=0.6002
p=0.0549
p=0.4254
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

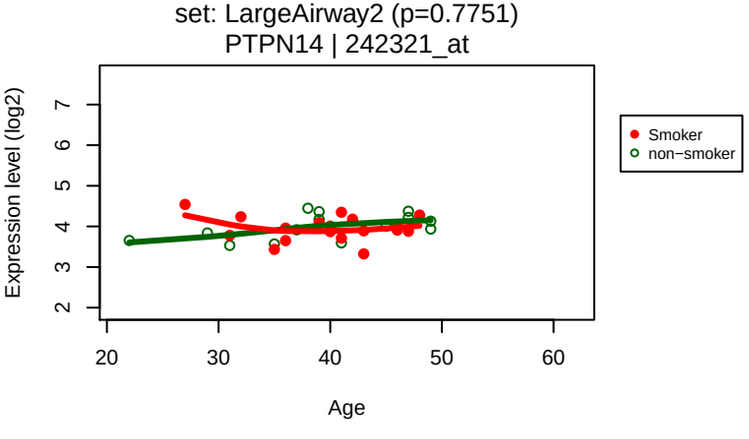

Gene expression ratio (smokers/nonsmokers)
gene: PTPN14 | 242321_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.822
p=0.899
p=0.5557
p=0.7193
p=0.7751
p=0.6002
p=0.0549
p=0.4254
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

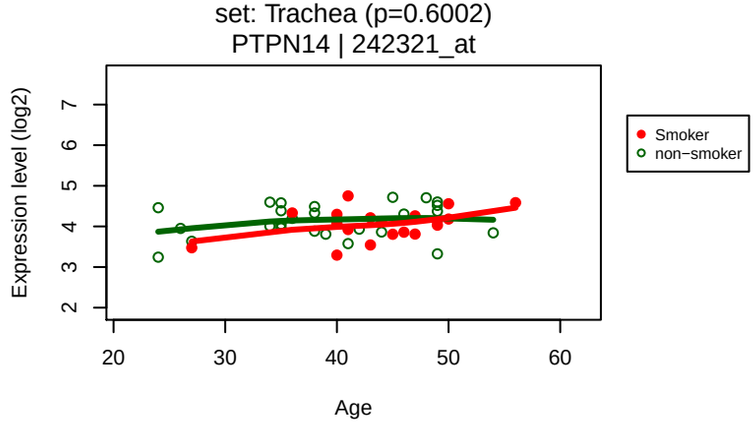

Gene expression ratio in Males
gene: PTPN14 | 242321_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7327
p=0.3895
p=0.3874
p=0.8833
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN14 | 242321_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5164
p=0.3031
p=0.5677
p=0.1544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

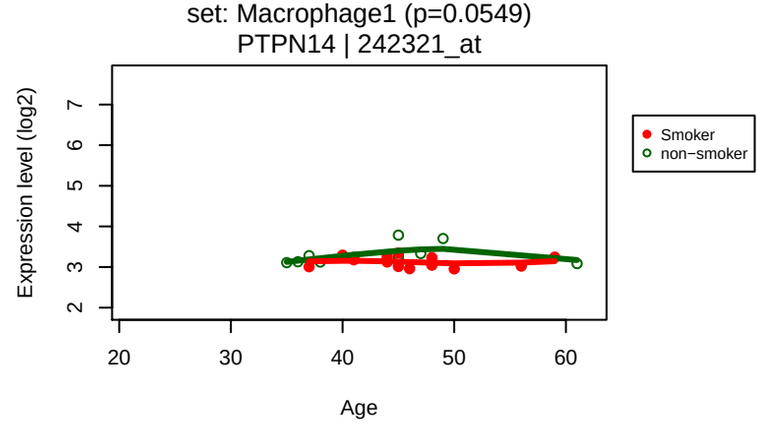

set: SmallAirway1 (p=0.822)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker

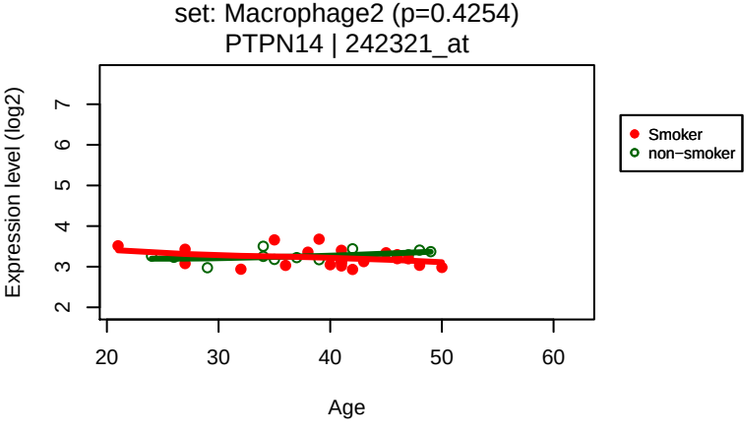

set: SmallAirway2 (p=0.899)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker

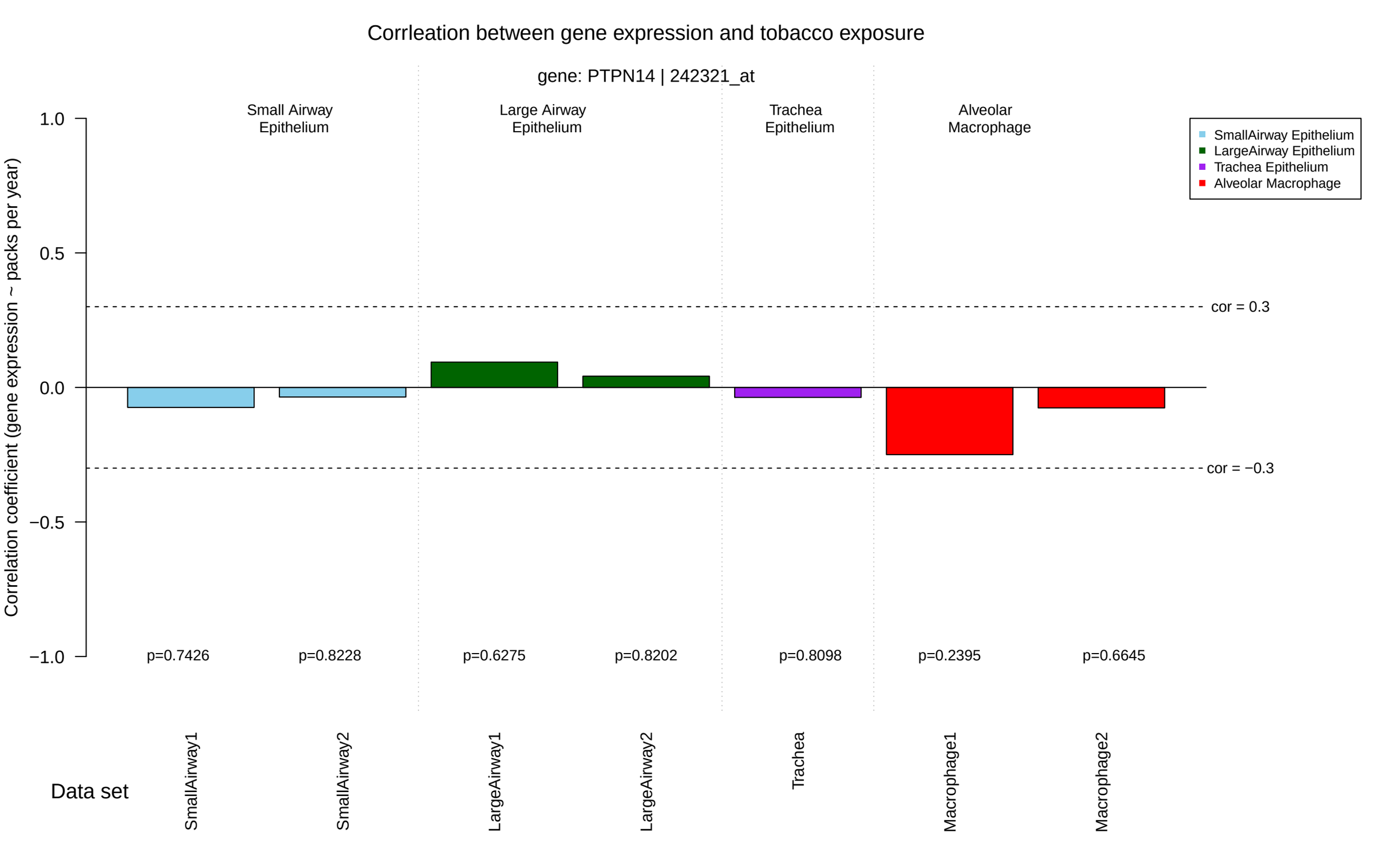

set: LargeAirway1 (p=0.7193)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7751)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6002)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0549)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4254)
PTPN14 | 242321_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN14 | 242321_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7426
p=0.8228
p=0.6275
p=0.8202
p=0.8098
p=0.2395
p=0.6645
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2